

Inflation, Employment Growth, and Housing Affordability in the United States (2016–2026):

An Analysis of Consumer Prices, Labor Market Trends, and Regional Rent Growth

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Motivation

Inflation has been one of the most significant economic challenges facing households in the United States over the past decade. Rising prices for essential goods and services reduce purchasing power and increase the cost of living, making it more difficult for individuals and families to maintain their standard of living. While inflation affects nearly every aspect of household spending, housing costs are often among the largest expenses faced by consumers and therefore play a critical role in overall affordability.

Housing affordability has become an increasingly important issue in recent years as rental prices have risen rapidly across many metropolitan areas. Reports from organizations such as the U.S. Bureau of Labor Statistics (BLS), Zillow, and various housing policy groups have documented substantial increases in both consumer prices and housing costs since 2020. At the same time, labor market conditions have changed considerably, with some industries experiencing strong employment growth while others have expanded more slowly. These trends raise important questions about whether rising costs have been accompanied by economic growth that improves household financial well-being.

Previous studies and public reports have frequently examined inflation, employment, and housing markets independently. However, fewer analyses investigate how these factors interact over time. Understanding the relationship between inflation, employment growth, and regional housing costs can provide valuable insight into broader affordability challenges faced by households across the United States.

This project combines data from the Bureau of Labor Statistics and Zillow to examine economic trends between 2016 and 2026. By analyzing changes in consumer prices, employment levels across major industries, and rent growth across metropolitan regions, we aim to better understand how affordability has evolved during a period characterized by both economic expansion and elevated inflation.

The primary research questions guiding this study are:

1. How have consumer prices changed between 2016 and 2026, and which categories experienced the largest increases?
2. Which industries experienced the greatest employment growth during this period?
3. How has rent growth varied across different regions of the United States?
4. Do regions experiencing stronger housing cost growth also exhibit different employment growth patterns?
5. What do these combined trends suggest about changes in housing affordability and overall cost-of-living pressures for U.S. households

Data Sources

Dataset 1: BLS Average Prices

- **Source:** U.S. Bureau of Labor Statistics
- **URL:** <https://data.bls.gov/toppicks?survey=ap>
- **Format:** Text file containing monthly average consumer prices for selected grocery and household items.
- **Variables:** The dataset includes a Series ID identifier along with monthly average price values for each item.
- **Time Period:** January 2016 – May 2026. Observations are used in full for this analysis.
- **Record Count:** The dataset contains 15 item observations and 134 variables.

Dataset 2: BLS Employment

- **Source:** U.S. Bureau of Labor Statistics
- **URL:** <https://www.bls.gov/webapps/legacy/cesbtab1.htm>
- **Labels URL:** <https://www.bls.gov/web/empsit/cesseriespub.htm>
- **Format:** Text file containing monthly employment counts by industry or sector.
- **Variables:** The dataset includes an encoded Series ID along with monthly employment count values for each sector.
- **Time Period:** January 2016 – May 2026. Observations are used in full for this analysis.
- **Record Count:** The dataset contains 348 sector observations and 133 variables.

Dataset 3: Zillow Observed Rent Index (ZORI)

- **Source:** Zillow Research Housing Data
- **URL:** <https://www.zillow.com/research/data/>
- **Format:** CSV file containing monthly rental index estimates for U.S. metropolitan areas.
- **Variables:** The dataset includes geographic identifiers such as RegionID, RegionName, RegionType, and StateName, along with monthly Zillow Observed Rent Index (ZORI) values. ZORI is a repeat-rent index designed to track changes in rental prices over time.

- **Time Period:** January 2015 – April 2026. For this analysis, observations beginning in January 2016 were used to align with the employment and inflation datasets.
- **Record Count:** The dataset contains 720 geographic observations and 141 variables. After filtering to metropolitan statistical areas (MSAs), the dataset was used to analyze regional housing trends across the Northeast, Midwest, South, and West regions of the United States.
- **Purpose in This Project:** The Zillow rent dataset serves as the primary measure of housing affordability. Metropolitan areas were grouped into four Census-style regions, and average regional rent indexes were calculated and normalized to January 2016 = 100. This allowed regional housing-cost trends to be compared directly over time and evaluated alongside national inflation and employment measures.

Data Manipulation Methods

The inflation data took a few steps to get into a usable shape. We started by pulling the labels into a text file and then mapping the Series IDs to them, so each row was tied to a real product like gasoline or eggs instead of just a code. The data also came in wide format, with every month spread across its own column, so we transformed it from wide to long so that time ran down the rows the way pandas wants it for any kind of time based analysis.

From there we handled the missing values, since the future 2026 months came in empty and some cells were blank strings rather than true NaNs, which would have broken the math later. Once everything was clean and numeric, we created an index by standardizing each series to January 2016 equals 100, so all the items sat on the same scale and could be compared by how much they grew rather than by their raw price.

The employment series data arrived in a wide layout, with each industry identified by a coded series ID and a separate column for every month. Preparation began by renaming the identifier column for consistency and reshaping the table from wide to long form, producing one row per series-month. Month labels formatted as text were parsed into proper dates so observations could be ordered and filtered chronologically. The employment values themselves required cleaning. Some carried a “P” preliminary tag, and trailing columns for future months were blank, so the tags were stripped, the field was coerced to numeric, and rows with missing dates or values were dropped. A flag distinguishing seasonally adjusted (CES) from unadjusted (CEU) series was added so the two could be separated cleanly, leaving a tidy long table spanning January 2016 through early 2026.

Additionally, the cleaned long table was narrowed to six seasonally adjusted series, each mapped from its cryptic code to a readable industry name. A year field was derived from the parsed dates, and employment was averaged within each industry-year to smooth seasonal swings. Percentage growth was then computed by comparing the 2025 annual average against the 2016 baseline for every industry, yielding a single comparable growth figure per sector. Sorting these values produced the ranked ordering used throughout, isolating the leading and lagging industries from a dataset that originally mixed dozens of overlapping series at differing levels of aggregation.

The Zillow Observed Rent Index (ZORI) dataset required several preprocessing steps before it could be used for regional housing affordability analysis. First, the raw dataset was imported from Zillow's metro-level rental data file and inspected to identify geographic and temporal variables. The dataset contained monthly rent estimates for hundreds of geographic areas, including both metropolitan statistical areas (MSAs) and a national-level observation.

Because the goal of this project was to compare regional housing markets, the dataset was filtered to retain only metropolitan statistical areas (MSAs). This removed the national aggregate observation and ensured that all remaining records represented local housing markets. Geographic information, including state abbreviations and metropolitan area names, was preserved for subsequent regional classification.

Next, each metropolitan area was assigned to one of four Census-style regions: Northeast, Midwest, South, or West. This classification was performed using state-level mappings, allowing metropolitan areas within the same region to be aggregated and compared. Regional grouping reduced geographic complexity while still capturing meaningful differences in housing market behavior across the United States.

After assigning regions, average monthly rent values were calculated for each region by aggregating all metropolitan areas within that region. To facilitate comparison across regions with different baseline rent levels, regional rent values were converted into an index with January 2016 set equal to 100. Indexing allowed relative rent growth to be compared on a common scale rather than comparing absolute dollar values.

The resulting regional rent index was used to evaluate housing cost growth over time and served as the primary measure of housing affordability trends within the project.

With both datasets cleaned, we combined them by lining them up on a common timeline so the months matched row for row. We normalized everything to January 2016 equals 100 so the two sources sat on the same scale, then ran year over year calculations to strip out the shared upward trend and the seasonality, which let us see whether things actually accelerated together rather than just both rising over time. From there we created a stress index that blended the price and employment sides into a single number we could track against housing.

Challenges Encountered

This project presented a few meaningful challenges. First, staying within scope was harder than expected with datasets these large and foreign. It was easy to envision countless visualizations, but narrowing them down to only those that directly served our research question required discipline. Additionally, decoding the encoded Series ID in the employment dataset was a challenge that was fun to try and decipher.

Working across three datasets that did not share a common unit also proved tricky. The employment data measured headcount by sector, while the rent and inflation datasets were

dollar-denominated. Reconciling those differences required figuring out a way to create a '*stress index*' utilizing the employment and inflation numbers to make meaningful comparisons across sources.

The biggest challenge for the industry growth notebook and other EDA was wrangling the raw BLS data into something usable. It comes in wide format with one column per month, so the first step was always reshaping it into long form. The values themselves needed work too: employment figures carried (P) preliminary tags, price data contained blanks and placeholder characters, and future months were simply empty. Each of these had to be stripped out, coerced to numeric, and dropped where missing. Date formats were also inconsistent between series, which required careful parsing to get everything aligned on a common timeline.

The series IDs were the toughest part, since a code like CES6562000001 is meaningless on its own and had to be manually mapped back to a readable industry name. The dataset also only included headcounts with no wage information, which kept the analysis focused on employment alone. Finally, because employment growth and price growth aren't directly comparable measures, inflation appears in the notebook as rough context rather than a true apples-to-apples comparison.

One challenge in the housing affordability analysis was that the Zillow dataset contained data at multiple geographic levels, including a national observation and hundreds of metropolitan areas. Determining an appropriate level of aggregation required balancing geographic detail with interpretability. Analyzing every metropolitan area individually would have produced overly complex visualizations, while relying only on national averages would have masked important regional differences. To address this issue, metropolitan areas were grouped into four Census-style regions, allowing meaningful geographic comparisons while maintaining a manageable level of complexity.

A second challenge involved comparing housing costs with the employment and inflation datasets. The Zillow dataset measured rent levels, while the other datasets measured employment counts and consumer prices. Because these variables were recorded in different units and scales, direct comparisons were not meaningful. To overcome this limitation, each series was normalized to a common January 2016 baseline, allowing trends to be compared through relative growth rather than absolute values.

Analysis 1: Inflation and Employment Relationships

We used year-over-year percent change instead of raw numbers because employment and prices both trend upward, so correlating the levels would just show everything moving together without meaning anything. Converting to YoY growth strips out that shared trend and the seasonality, so the correlation actually reflects whether two things accelerate together.

The employment data needed a few cleanup steps before we could analyze it. First we stripped out the "(P)" markers, which just flag preliminary values in the BLS data, since they'd otherwise get in the way of the actual numbers. Then we converted those columns to numeric

so we could run calculations on them instead of pandas treating them as text. From there we mapped each record to its industry and decoded the Series IDs, which have the industry code and the seasonal adjustment indicator baked right into them. Last, we pulled only the seasonally adjusted series, so the employment numbers show the real underlying trend instead of the predictable seasonal swings that happen every year.

The clearest take away is that there is a strong correlation between a rise in energy prices, fuel and oil, and an increase in hiring. This does not mean that rising fuel prices causes jobs to be created. We believe it is more along the lines of increased fuel demand and increase in hiring are both commonalities in a hot economy. When there is a change in the supply/demand ratio there will be a shift in price either up or down. So if demand goes up it is assumed the price will also increase.

Analysis 2: Regional Housing Affordability

To examine regional differences in housing affordability, metropolitan areas from the Zillow Observed Rent Index dataset were grouped into the Northeast, Midwest, South, and West regions. Average monthly rent values were calculated for each region and converted into an index with January 2016 set equal to 100, allowing rent growth to be compared despite differences in baseline rent levels.

The analysis revealed that rent growth was not uniform across the United States. While all four regions experienced substantial increases in housing costs between 2016 and 2026, the West exhibited the strongest overall rent growth, followed by the South. Both regions experienced particularly rapid increases following the COVID-19 pandemic, suggesting heightened housing demand and affordability pressures. In contrast, the Midwest and Northeast experienced more moderate rent growth throughout much of the study period.

These findings indicate that housing affordability challenges were highly dependent on geographic location. Although housing costs increased nationwide, residents of the West and South experienced larger relative increases than those in the Northeast and Midwest. When considered alongside the project's inflation and employment analyses, the results suggest that affordability pressures were driven not only by rising consumer prices but also by substantial regional differences in housing costs.

Analysis 3: Industry Growth Compared to Inflation

Between 2016 and 2025, employment growth across major industries was strikingly uneven. Construction, Healthcare and Professional Services were the clear leaders, each expanding its workforce by roughly a fifth as inflation grew. This was a sizable gap compared to the other industries: Leisure & Hospitality, Manufacturing, and Retail. Because Healthcare grew from an already large base of about 18.5 million jobs to 23.8 million jobs, its percentage gain also represents the largest absolute increase in headcount of any industry examined.

A broad service vs. goods divide underlies these results. The fastest growing sectors were service oriented, while the two goods heavy categories, Manufacturing and Retail, lagged furthest behind, consistent with longer run shifts towards automation and e-commerce. Set against the inflationary backdrop, the contrast sharpens further: consumer prices in the sampled basket rose roughly 45% over the same window, far outpacing employment gains in every industry. Even the strongest job growth fell well short of the cost-of-living increase, underscoring that the period's expansion in employment was modest relative to the scale of price growth, though the two measures track fundamentally different quantities and are best read as context rather than a direct comparison.

Visualizations

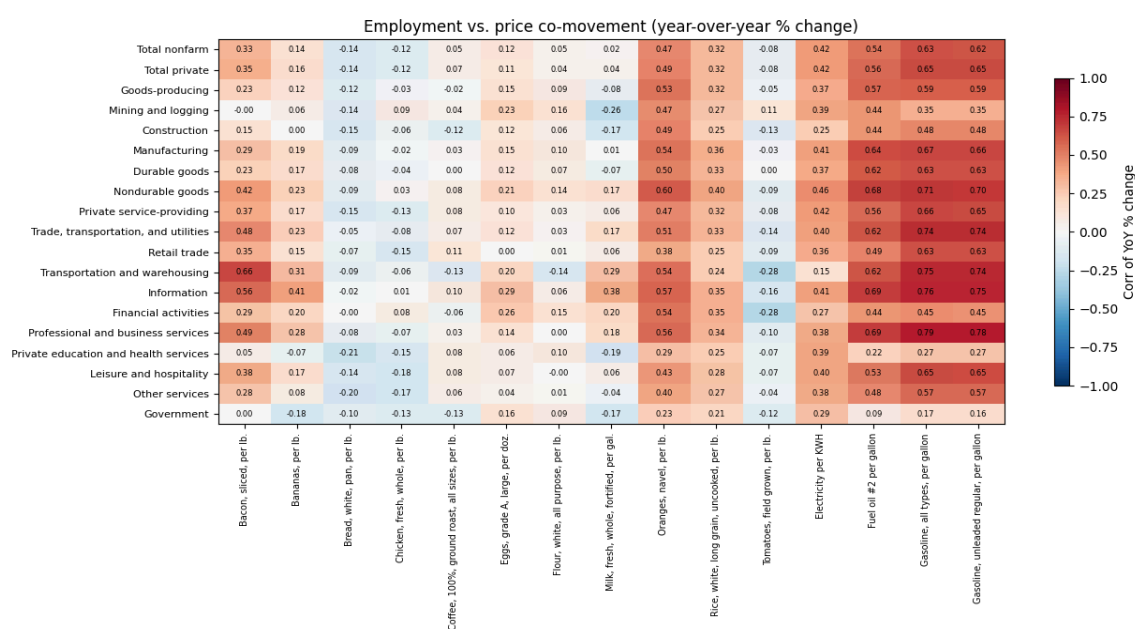


Figure 1. Heatmap showing correlation between job growth and inflation throughout the sectors.

A heatmap was the right pick for this part of the analysis because what we were looking at is a grid of relationships, not a single neat pairing. With roughly 18 industries measured against 15 price items, there were far too many comparisons for a line chart, scatter plot, or histogram to show cleanly or efficiently. A heatmap was the only format that could hold all of those relationships in one view at once.

It also fit the data itself. The correlations form a two dimensional matrix that maps cleanly onto a diverging color scale, with both axes categorical and the value continuous. That made the heatmap the most legible choice by far, since color does the work of showing strength and direction without cluttering the picture.

The heatmap surfaces three main things. Fuel and gas prices co-move with almost every sector, packaged food prices don't really track employment at all, and transportation and

warehousing tracking with fuel stands out as one of the most trustworthy signals because there's a real mechanism behind it. Taken together, the picture suggests that rising fuel and gas prices reflect an increase in demand, which makes them a good thermometer for the health of the economy as a whole.

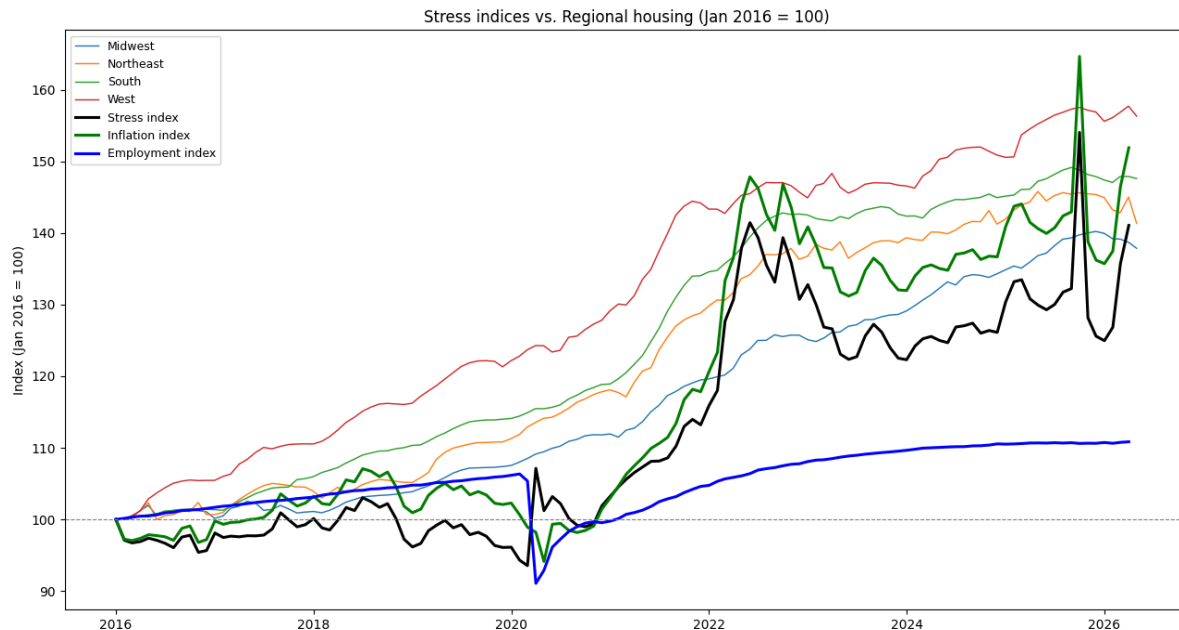


Figure 2. Stress Index vs. Regional Housing (January 2016 = 100)

This visualization was chosen to compare regional housing cost trends with broader economic indicators over a common time period. The figure combines regional rent indexes for the Northeast, Midwest, South, and West with national inflation, employment, and housing stress indexes, all normalized to a January 2016 baseline of 100. Using indexed values allows variables measured in different units to be displayed on the same scale and compared directly.

The visualization is appropriate because the project's primary objective is to examine housing affordability within the broader context of inflation and employment trends. A multi-line time-series chart effectively highlights changes over time while allowing differences between regions to be observed simultaneously. By displaying all series on a common index scale, the figure makes it possible to identify periods when housing costs grew more rapidly than employment or inflation.

The figure shows that rent growth was not uniform across the United States. The West experienced the largest overall increase in housing costs during the study period, while the South also exhibited substantial growth. In contrast, the Midwest and Northeast experienced more moderate increases. The chart also illustrates that regional rent growth generally exceeded employment growth and, in some periods, closely tracked or surpassed broader inflationary pressures. These patterns suggest that housing affordability challenges were

influenced not only by rising consumer prices but also by significant regional differences in housing market conditions.

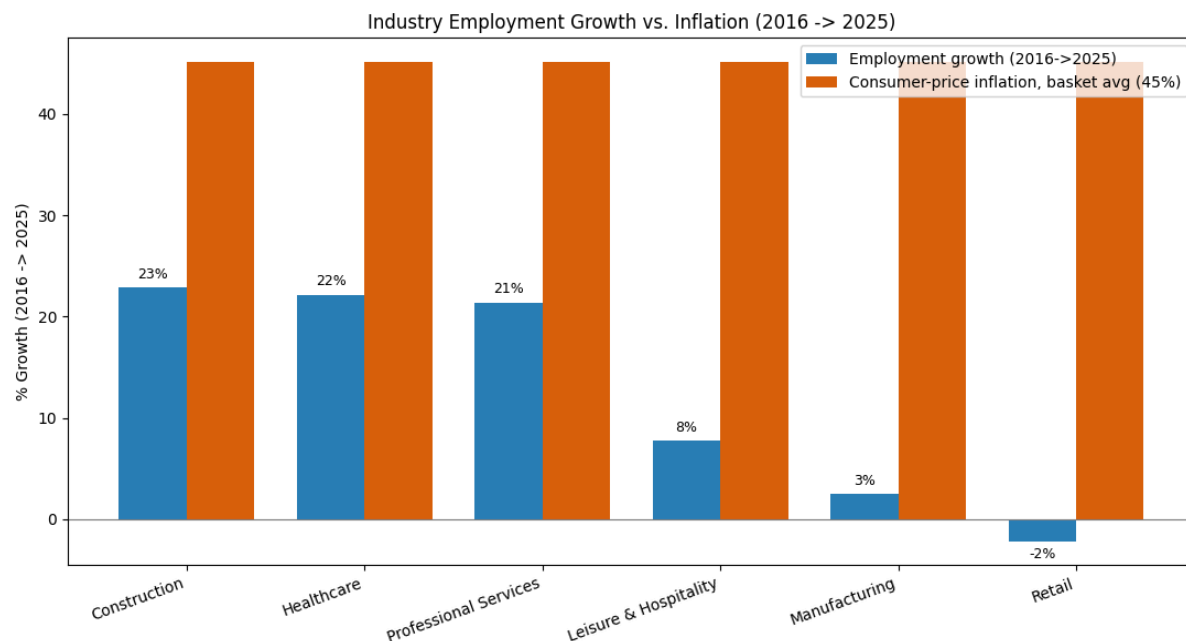


Figure 3. Industry Employment Growth vs. Inflation (2016 - 2025)

The figure is a grouped bar chart comparing how much employment grew across six U.S. industries against the rise in consumer prices over the same period. Each industry shows a blue bar for its employment growth alongside an orange bar marking at roughly 45% average inflation benchmark. Construction (+23%), Healthcare (+22%), and Professional Services (+21%) saw significant growth. Leisure & Hospitality and Manufacturing trailed well behind, and Retail was the only sector to shrink. All of them grew far slower than prices rose.

Conclusion

This project examined how inflation, employment growth, and housing costs changed across the United States between 2016 and 2026. The results showed that consumer prices increased substantially during the study period, with energy-related categories generally exhibiting some of the strongest relationships with economic activity. Employment growth varied considerably across industries, with sectors such as transportation, construction, and healthcare experiencing stronger growth than others. Housing costs also increased across all regions of the United States, although rent growth was most pronounced in the West and South.

When the datasets were considered together, the findings suggested that affordability pressures were driven by a combination of rising consumer prices and increasing housing costs. While employment generally expanded during the study period, growth was uneven across

industries and did not necessarily correspond to the regions experiencing the largest increases in rent. These results indicate that cost-of-living challenges were not distributed uniformly across the country.

Several limitations should be acknowledged. The employment dataset measured employment counts rather than wages, limiting direct comparisons between income growth and housing costs. Additionally, the project combined national employment and inflation measures with regional housing data, which may mask local economic differences.

Future research could incorporate wage or income data, investigate affordability at the metropolitan level, and explore additional economic factors that influence housing costs. Such analyses would provide a more comprehensive understanding of how inflation, employment, and housing markets interact to affect household affordability.

Statement of Work

Colten Wasden prepared and analyzed the Bureau of Labor Statistics (BLS) Average Price dataset. He identified and mapped BLS series identifiers to consumer-price categories, cleaned and transformed the inflation data, and developed the inflation index used throughout the project. He designed and implemented the employment-versus-inflation correlation analysis, calculated year-over-year changes, created the correlation heatmap visualization, and contributed the inflation-related portions of the report. Colten also assisted with integrating the datasets into a common framework for comparison and participated in project planning discussions.

Gabriella Gass identified, prepared, and analyzed the Zillow Observed Rent Index (ZORI) dataset. She cleaned and filtered the metro-level rental data, grouped metropolitan areas into the Northeast, Midwest, South, and West regions, and created indexed rent measures with January 2016 set equal to 100. She performed the regional housing affordability analysis, developed the regional housing visualization, and wrote the housing-related portions of the report, including sections covering the Zillow data source, rent data preparation methods, housing affordability analysis, visualization explanation, and conclusions. Gabriella also coordinated team communication, organized report sections, scheduled meetings with the project advisor, and maintained the shared project document.

Garret Obenchain prepared and analyzed the BLS employment dataset. He cleaned the employment data, removed preliminary value indicators, converted employment counts into a usable format, and organized industries into meaningful categories for analysis. Garret examined long-term employment growth trends across major sectors and created the industry employment growth visualization comparing sector growth against cumulative inflation. He contributed the employment-related portions of the report, including data preparation methods, analysis, visualization interpretation, and conclusions.

The team collaborated through Discord, GitHub, Google Docs, and meetings with our project advisor. Major decisions regarding datasets, research questions, analytical methods, and visualizations were made collectively. Each team member was responsible for one primary dataset and analysis component while regularly sharing progress, reviewing each other's work, and coordinating report development.

Future collaboration could be improved by integrating the three analyses earlier in the project and establishing shared metrics sooner. Additional integration between the employment, inflation, and housing components may have produced even stronger cross-dataset insights. Future work could also incorporate wage or income data to provide a more direct measure of affordability.